

# japa 494 log lifter

## User manual

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## **1. Introduction**

TP Silva Oy (JAPA) is a Finnish company engaged in extensive product development. Our goal is to produce simple, reliable and structurally durable machines with a long service life. If you use your JAPA machine correctly and maintain it according to the instructions in this manual, it will serve you efficiently for a long time. If you have questions and would like to discuss them in detail, please contact our retailer or us directly.

### **CONGRATULATIONS ON PURCHASING A JAPA LOG LIFTER!**

This manual is intended for operators with the appropriate expertise. For this reason, operators are expected to have the general knowledge and basic skills required for working with this kind of machinery. Familiarise yourself with the manual before installing the machine and beginning operation. Carefully study the machine's features and safety equipment before operation. Always keep this manual with the machine. The instructions, descriptions and technical specifications in this manual are based on the latest information with regard to the workings of the machine at time of writing. However, the manufacturer develops and updates the machine continuously and therefore reserves the right to change the machine's features and safety-related properties without separate notification.

In order to receive quick and efficient assistance when ordering spare parts and encountering any possible fault situations, you must indicate the information on the machine's type plate to the retailer or service technician. Write down the type plate information in the reserved section on this page to ensure that it is readily available at all times. If you cannot find a solution to a problem on your own, please contact the retailer, who will then resolve the matter with the manufacturer.

We at JAPA are confident that you will be satisfied with your new firewood processor. It meets all the safety requirements imposed by the European Union and carries the relevant CE label.

### **1.1 Customer registration**

TP Silva Oy utilises an EXTRANET service where owners and operators can register their machines. The site contains useful information, such as user manuals and spare parts manuals.

**<https://info.japa.fi/>**



**FILL IN THE TYPE PLATE INFORMATION AND THE RETAILER'S CONTACT INFORMATION:**

Serial number: \_\_\_\_\_

Delivery date: \_\_\_\_\_

Retailer: \_\_\_\_\_

Contact person: \_\_\_\_\_

Address: \_\_\_\_\_

Telephone: \_\_\_\_\_

## **1.2 Declaration of incorporation for partly completed machinery**

**Manufacturer:**

TP Silva Oy  
Kusnintie 44  
FI-23800 Laitila, Finland  
Tel. +358 2857 1200  
Fax +358 2857 1201  
Web: [www.japa.fi](http://www.japa.fi)

Person responsible for the technical file: Ville Kairus

**The declaration applies to the following machines:**

JAPA 494 log lifter, firewood processor accessory

This partly completed machine may only be connected to a suitable Japa firewood processor that meets the requirements of the Machinery Directive.

**Drive:**

Auxiliary hydraulics valve of the firewood processor

**The following directives have been observed in the construction of each machine:**

Machinery Safety Directive 2006/42/EC enacted through Government Decree 400/2008.

Upon request with the proper grounds, we undertake to supply the necessary technical documents concerning partly completed products to national authorities.

Laitila 1 October 2020

**TP Silva Oy**



**Jori Lammi**  
Managing Director



### 1.3 Purpose of use

The JAPA log lifter is a safe and easy-to-use device for lifting logs from the ground onto the input table.

### 1.4 Machine models

Japa 494

Maximum log weight 200 kg



**ONLY ONE PERSON AT A TIME MAY OPERATE THE MACHINE!**

### 1.5 Instruction and warning labels on the machine



Maximum weight of lifted logs  
200 kg



Only one user at a time. Ensure  
that no other persons are in the  
working area.



Risk of crushing! Stay clear  
of the machine's moving parts.



Read the user manual  
before operation and  
maintenance.



Check the condition and  
safety equipment of the machine before  
operation.



**DO NOT LEAVE A RUNNING MACHINE UNSUPERVISED!**

## 1.6 The machine's type plate

The machine's type plate is located behind the machine, on the tractor side.

|                                                                                                                |  |                                                                                   |
|----------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------|
| Sarjanro                                                                                                       |  |  |
| Manufact.Nr                                                                                                    |  |                                                                                   |
| Malli, Model                                                                                                   |  |                                                                                   |
| Pvm, Date                                                                                                      |  |                                                                                   |
| Paino, Weight                                                                                                  |  |                                                                                   |
| Jännite, Voltage                                                                                               |  |                                                                                   |
| R/min                                                                                                          |  |                                                                                   |
| Hydr.max paine                                                                                                 |  |                                                                                   |
| Terä, Blade                                                                                                    |  |                                                                                   |
| <b>JAPA - MACHINE</b><br><b>LAITILAN RAUTARAKENNE OY</b><br><small>www.japa.fi, (02) 8571 200, FINLAND</small> |  |                                                                                   |

Serial number  
Machine type  
Production year and date  
Weight of the machine  
Voltage (E and TRE models)  
Maximum RPM of the power take-off  
Maximum hydraulic pressure  
Diameter of the cutting blade/blade opening  
Name and address of manufacturer

## 1.7 Safety instructions

These safety instructions are general in nature. When handling the machine, you must take into account all other relevant instructions concerning health and safety, road traffic regulations regarding transportation, as well as the provisions of general legislation. Observing these instructions helps to prevent accidents.

The machine may only be used by persons who have familiarised themselves with the manual and the operation of the machine. The operator must not be under the influence of alcohol or narcotic substances. The requirements placed on the operator, as well as the minimum age of 18 required to operate the machine, must be taken into account.

Study the machine's safety and installation instructions carefully, along with the operation and control functions, before installing and operating the machine.

### General provisions:

- The machine must be used solely for lifting logs.
- Only one person may operate the assembly.
- Ensure that no unauthorised persons are in the working area. The danger zone is 10 m.
- Always check the condition of any possible power cables.
- Always use approved eye and ear protection.
- Clean and maintain the machine on a regular basis.
- Always stop the machine before maintenance and, if the machine is tractor-driven (TR), disconnect the cardan shaft. If the machine features an electrical drive (E), disconnect the power cable.
- Check the operation of the safety equipment. Do not use the machine if part of the mechanism is not functioning properly.
- The machine may only be transported in the transport position. This is the responsibility of the driver.

### Working area:

- Select a level and solid working area.
- Keep the area clean and free of obstructions.
- Ensure that the ground or floor is not slippery near the machine.
- Do not use the machine indoors, as the dust may enter your airways or cause a fire hazard.
- Only use the machine with sufficient lighting: daylight is recommended.
- Place the machine in the working position and check the safety devices always before starting the machine.

### During operation:

- Take extreme care when lifting knotty or twisted wood.

- When lifting incorrectly, the log may spin or fall, causing injury or machine damage.
- Careless lifting may cause unexpected hazards.

## 1.8 Guarantee terms

### Roles

#### **Guarantor**

Manufacturer of the JAPA product:

TP Silva Oy, Kusunntie 44, FI-23800 LAITILA

Tel.: +358-(0)2-857 1200, Fax: +358-2-857 1201, Email: aftersales@japa.fi.

#### **Retailer**

The retailer is a company authorised by TP Silva Oy, which sells and markets JAPA products in its local area. The retailer acts as the recipient in warranty matters regarding JAPA products it has sold to the buyer.

#### **Buyer**

The buyer is the person or community that acquires a JAPA product for personal use. The buyer is obliged to report faults within the warranty terms to the retailer and to retain the receipt in order to prove where and when the JAPA product has been purchased. When necessary, the buyer is also obliged to indicate the type plate information to the retailer.

#### **Operator**

The operator is the person or community that uses the machine.

The guarantee is valid for the original buyer for 12 months, starting from the date of purchase, but for no more than 1,000 operating hours.

In guarantee matters, always contact the machine's seller before undertaking any procedures.

A guarantee claim must be issued to the seller in writing **immediately** upon discovery of a defect. If the defect concerns a damaged part or component, please send a photograph of the damaged part or component to the seller, if possible, so the fault can be identified. When submitting a guarantee claim, the buyer must always include the type and serial number of the machine in the claim and present a receipt that includes the date of purchase. Guarantee claims must be submitted to an authorised retailer.

### **The guarantee covers**

- Parts damaged in normal use due to faults in the material or workmanship.
- Reasonable repair expenses in accordance with the agreement between the seller or buyer and the manufacturer. Faulty parts will be replaced with new ones. A faulty part or parts replaced due to a material fault must be returned to the manufacturer via the retailer.

### **The guarantee does not cover**

- Damage caused by normal wear and tear (such as blades and belts), improper use or use contrary to the instruction manual.
- Damage caused by negligence of maintenance or storage procedures detailed in the instruction manual.
- Damage caused during transport.
- Cutting blades, V-belts and oil as well as normal adjustment, care, maintenance or cleaning procedures.
- Defects in a machine to which the buyer has carried out or commissioned structural or functional changes, to the degree that the machine can no longer be considered equivalent to the original machine.
- Other potential costs or financial obligations resulting from the procedures mentioned above.
- Indirect costs.
- Travel costs resulting from guarantee repairs.
- The guarantee for parts replaced during the guarantee period of the machine expires at the same time as the machine's guarantee.
- The guarantee is void if the ownership of the machine is transferred to a third party during the guarantee period.
- The guarantee is void if any of the machine's seals have been broken.

If a fault or defect reported by the customer is found not to be covered by the guarantee, the manufacturer has the right to charge the customer for the troubleshooting and possible repair of the fault or defect in accordance with the manufacturer's current price list.

This guarantee certificate indicates our responsibilities and obligations in full and excludes all other responsibilities.

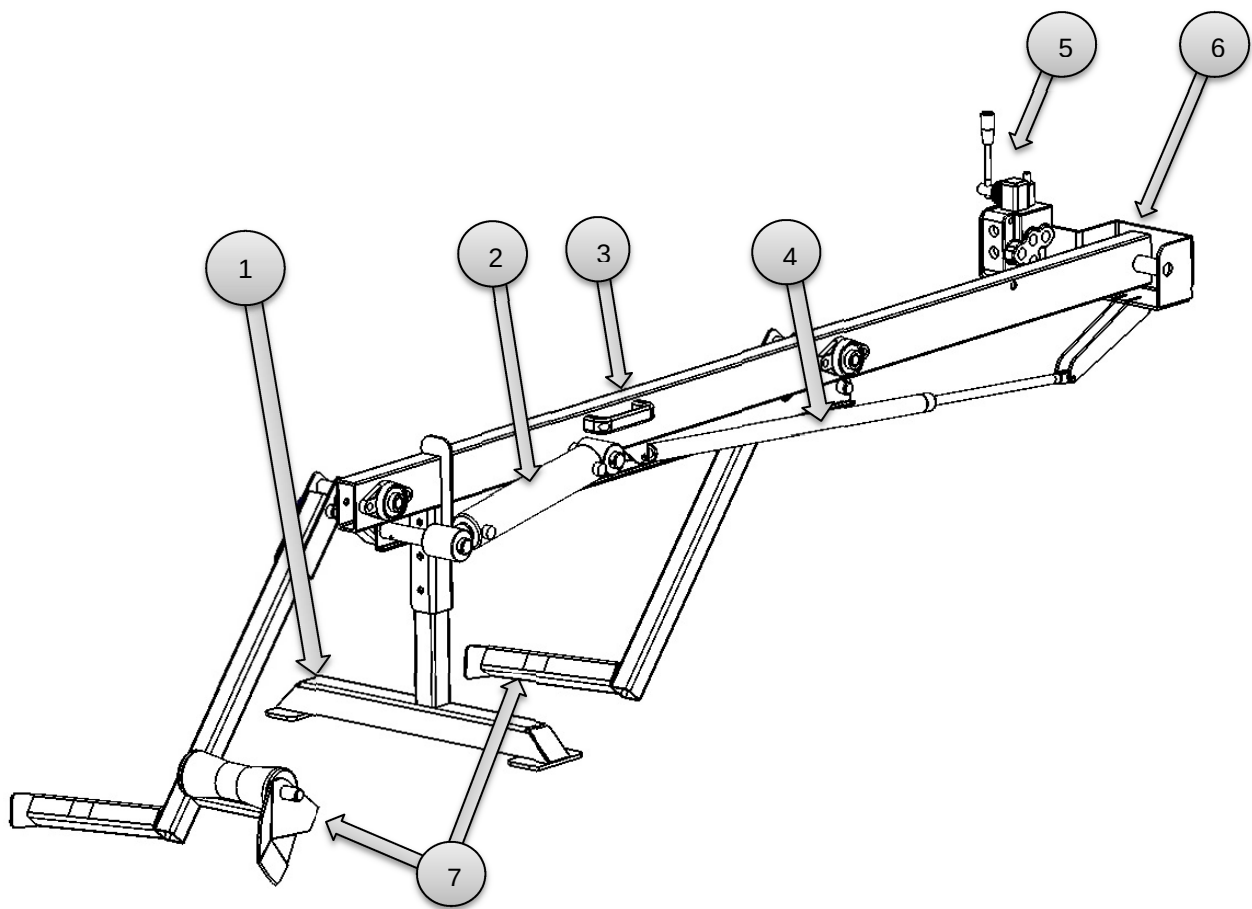
Guarantee terms come into force when you register your customership in the extranet service found on our website.

## **2. Installation of the machine**

### **2.1 Delivery inspection**

Inspect the machine immediately upon delivery. If the product has sustained damage in transit or it is missing components, contact the transport company and retailer immediately.

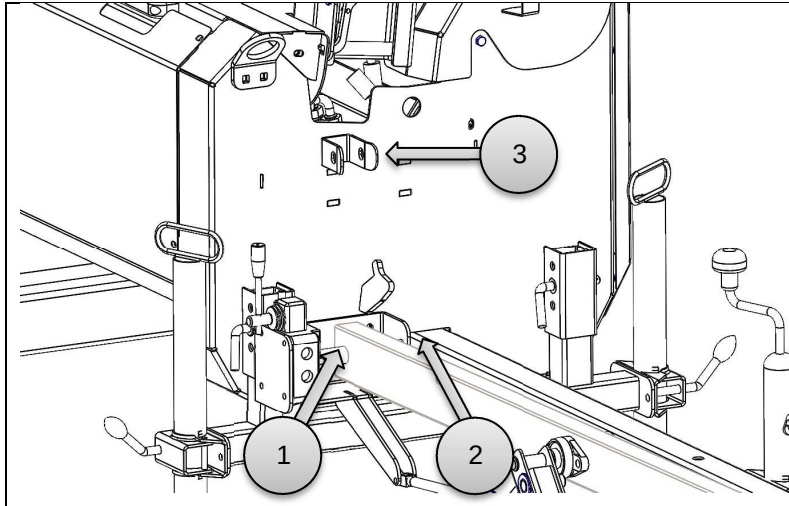
### **2.2 Main components of the machine**



1. Support leg
2. Lift cylinder
3. Lifting handle
4. Gas spring
5. Control valve
6. Lifter's mounting adapter
7. Lifting hooks

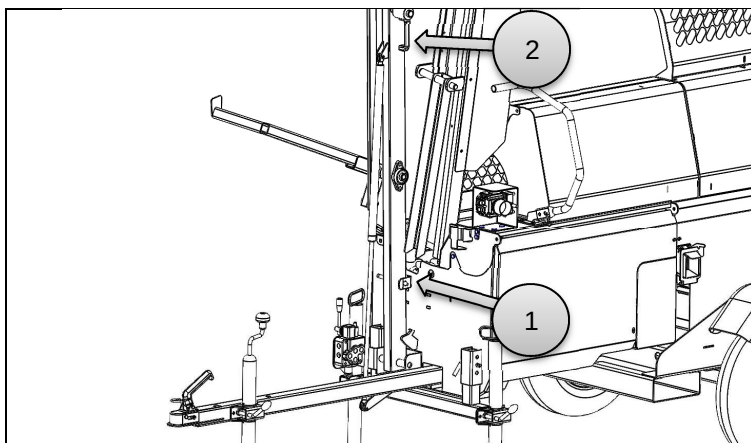


### 2.3 Attaching the log lifter to the machine

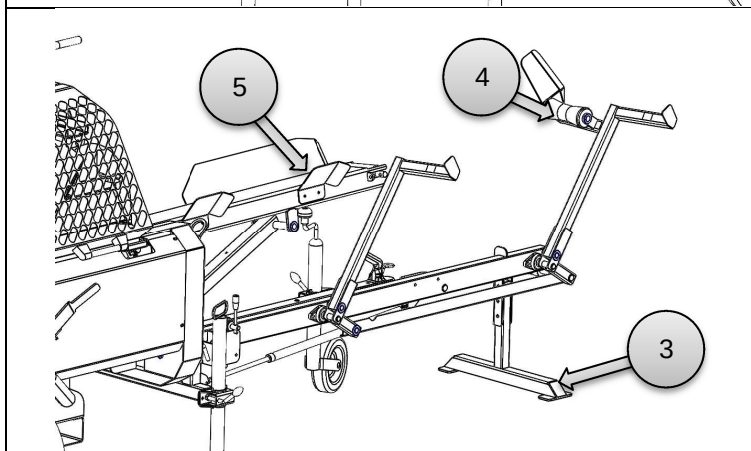


1. Attach the lifter's mounting adapter to the frame of the log lifter with the supplied M10 bolts (3 pcs).
2. Fasten the lifter to the adapter with a hinge pin (2) and lock it in place with a lock pin.
3. Attach the transport support to the firewood processor's frame.
4. Connect the hydraulic hoses to the processor's hydraulic system in accordance with the hydraulic schematic.

### 2.4 Placing the log lifter in the working position



1. Remove the transport support pin (1).
2. Grip the lifter handle firmly and lower the lifter to the ground. The lifter features a gas spring mechanism which makes the device easy to lift and lower.



1. Adjust the support (3) to the correct height so that the lifter is level.
  2. Ensure that the feed roller (4) is aligned and at the same height with the input conveyor when the lifting hooks are raised to the upper position.
- If necessary, detach the input conveyor's outermost log guide plate (5).

### 2.5 Placing the log lifter in the transport position

The machine can be placed in the transport position by performing the steps in section 2.4 in reverse.



**READ THE USER MANUAL BEFORE ACTIVATING THE MACHINE FOR THE FIRST TIME!**

### **3. Operating the machine**

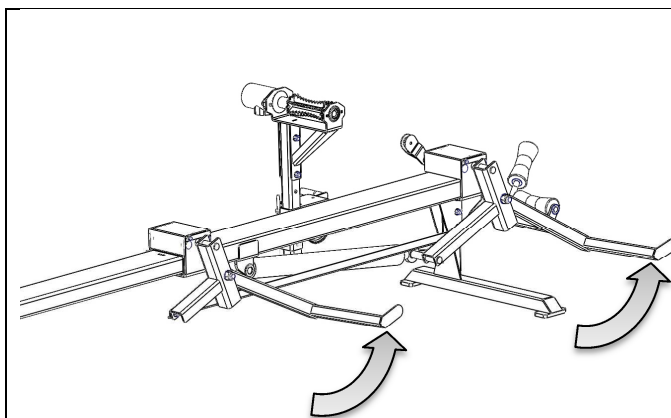
#### **3.1 Acknowledging the hazards and responsibilities involved**

Operators of the machine must familiarise themselves with and observe the instructions in the user manual. Incorrect operation of the machine (such as modifying the operation of the machine) may compromise the operator's safety. In these cases, the responsibility for the consequences rests with the operator.

#### **3.2 Before use**

Safe use of the machine requires adherence to the user manual and the instructions provided by the instruction and warning labels. The operator is responsible for the operation and maintenance of the machine and its safety equipment. Neglecting to maintain the machine or use the safety equipment will void the guarantee. Ensure that you have familiarised yourself with the operation of the machine and its functions.

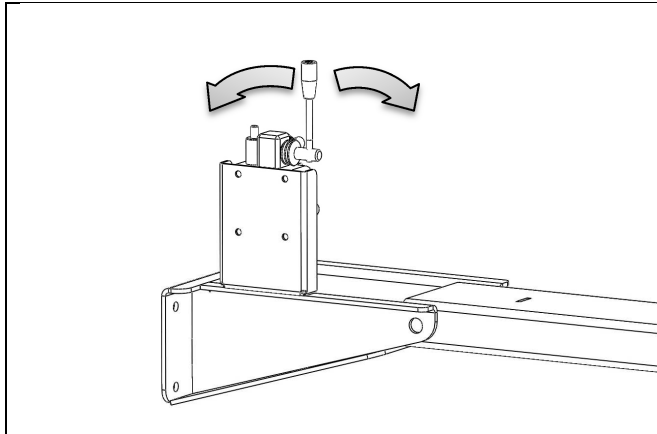
1. Visually check that the machine has not been damaged.
2. Ensure that the machine has been installed firmly in place.
3. Check the machine for oil leaks.
4. Ensure that the machine's power source is correctly connected.
5. Ensure that all of the machine's components and guards are in place.
6. Ensure that no unauthorised persons are in the danger zone.
7. Start the machine.
8. Check that the machine's controls work as intended.



Check that the safety mechanism of the lifting hooks works.

The lifting hooks must give way if met with an obstruction during lowering.

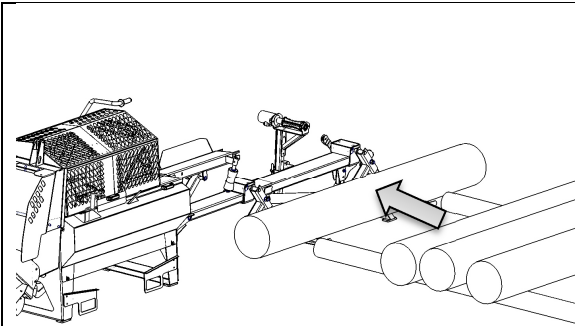
### 3.3 Operating the machine



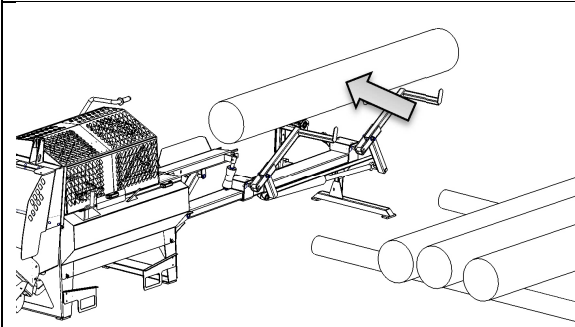
The lifting hooks move upwards when the valve handle is turned to the left.

The lifting hooks move downwards when the valve handle is turned to the right.

The lifting hooks remain in place when the valve is in the middle position.



Lower the lifting hooks to the lower position using the hydraulics control valve and roll the log onto the lifting hooks.



Lift the lifting hooks up using the hydraulics control valve and roll the log onto the feed rollers and input table.



**ENSURE THAT THE LOG WILL STAY FIRMLY ON THE LIFTING HOOKS FOR THE DURATION OF THE LIFTING AND CANNOT FALL OFF!**



**NOTE THE 200 KG WEIGHT LIMIT OF THE LOGS!**

## **4. Maintenance**

### **4.1 Hydraulics**

After connection, the hydraulic system's pressure is high. A fault under high pressure may cause the hydraulic oil discharging from the system to penetrate the skin and cause severe injury. The risk of injury is also present when inspecting the system for leaks. In the event of injury, contact a doctor immediately (risk of infection).

Always take extra care when working with hydraulic components. Use the necessary protective equipment and tools.

Check the hydraulic hoses at regular intervals and replace them if they are worn or damaged. All replacement hoses must comply with DIN standards or corresponding requirements.

#### **Note the following when replacing hoses**

- Always ensure that the hose is pressure-free before opening the connector.
- When replacing hoses, ensure that the hoses are not twisted and will not be twisted during the machine's operation.
- Do not overtighten the connector nut.
- Check the position of the connector.
- Use correctly sized hose ties.
- Tighten the threaded (=male) connectors before tightening the connector nuts.
- Use locking and sealing agents.
- Use correctly-sized spanners.

#### **Remove a hose and connector from use in the following cases:**

- The hose has come off the connector (also discard the connector)
- The hose has a tear or leak (also when the hose is moist due to internal seepage)
- The surface layer has worn all the way down to the support weave
- The hose has a permanently flattened or otherwise deformed section
- The hose displays signs of burning or the surface is cracked
- The hose manufacturer does not guarantee a life span exceeding five years
- The connector has corroded or features a permanent deformity
- The connector leaks
- The compression connector can only be used once

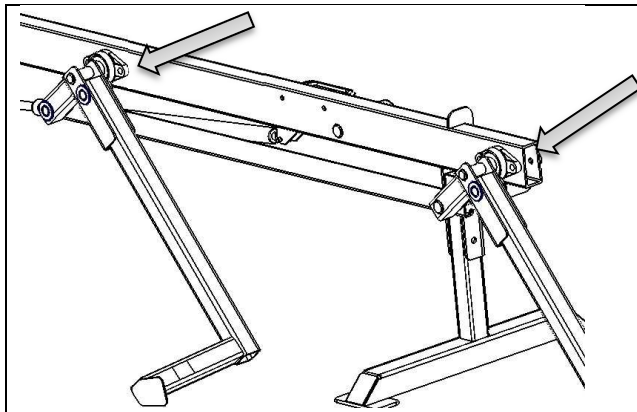
#### **Protecting yourself from oil and grease**

Oil and grease may contain additives that may, in some cases, be harmful with repeated skin contact. Adhere to the appropriate safety instructions when handling these substances.

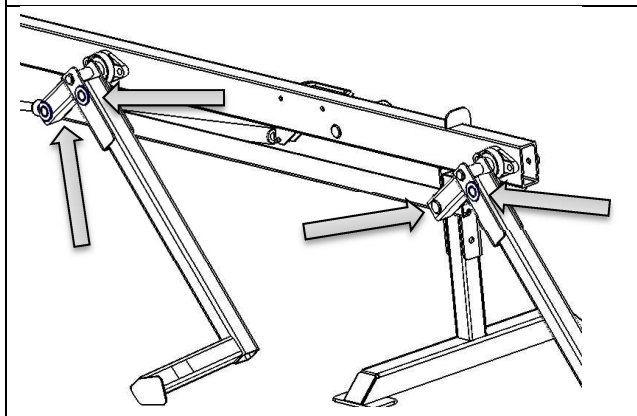
- When handling oil or grease, always wear the appropriate protective clothing and oil-proof gloves, along with using protective hand creams.
- The spreading of oil mist in the working area or the spilling of oil on pathways will cause a slipping hazard.
- Avoid skin contact with oil or grease.
- Never use oil or lubrication grease to clean your hands.
- Change oily clothes for clean ones as quickly as possible.
- Do not leave oily cotton or similar items in your pockets.
- Waste oil must be collected and disposed of according to the appropriate regulations.

## 4.2 Lubrication points

The bearings are self-lubricated and do not necessarily require lubrication. However, you may be able to extend the life span of a bearing by adding a small amount of lubricant at the end of the operating season. Excess lubricant in a bearing may damage the seal.



Inject a couple of doses of lubricant into each of the four bearings with a grease pump.



Lubricate all pivoting joints with a sprayable lubricating oil, for example.

## **5. Technical specifications**

| <b>Property</b>            | <b>JAPA 494</b> |
|----------------------------|-----------------|
| Max log diameter           | 31 cm           |
| Max log weight             | 200 kg          |
| Max log length             | 3 m             |
| Lifter weight              | 45 kg           |
| Maximum hydraulic pressure | 215 bar         |
| Maximum hydraulic output   | 37 litre/min    |

